

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

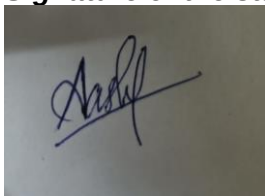
APPLICATION- BASED QUESTIONS

Code of Conduct:

I Aashika S P (192521068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a white background. The signature is stylized and appears to read 'Aashika'.

APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security** : A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Introduction

A student organization needs to securely communicate confidential event details with its core members. For this purpose, a simple classical encryption technique based on a fixed shift is used. The given plaintext message is “**MEET AT NOON**”, and a shift of **4 positions** is applied. This technique is known as the **Caesar Cipher**.

Encryption of the Message

The Caesar Cipher shifts every alphabetic character by a fixed number of positions.

The encryption formula is:

$$C=(P+K)\text{mod}26$$

Where:

- PP = numerical value of the plaintext
- CC = numerical value of the ciphertext
- KK = shift key
- $K=4$
- A = 0, B = 1, ..., Z = 25

The plaintext is: **MEET AT NOON**

Plaintext	Value	Formula	Ciphertext
M	12	$(12 + 4) \text{ mod } 26 = 16$	Q
E	4	$(4 + 4) \text{ mod } 26 = 8$	I
E	4	$(4 + 4) \text{ mod } 26 = 8$	I
T	19	$(19 + 4) \text{ mod } 26 = 23$	X
A	0	$(0 + 4) \text{ mod } 26 = 4$	E
T	19	$(19 + 4) \text{ mod } 26 = 23$	X
N	13	$(13 + 4) \text{ mod } 26 = 17$	R
O	14	$(14 + 4) \text{ mod } 26 = 18$	S
O	14	$(14 + 4) \text{ mod } 26 = 18$	S
N	13	$(13 + 4) \text{ mod } 26 = 17$	S

The encryption is performed as follows:

The spaces between words remain unchanged.

Therefore, the encrypted message is:

MEET AT NOON → QIIX EX RSSR

Ciphertext: QIIX EX RSSR

b. Decryption of the Message

An authorized member who knows the encryption rule and the key can retrieve the original message by shifting every ciphertext character **4 positions backward**.

The decryption formula is:

$$P=(C-K)\bmod 26$$

Since the key is 4:

$$P=(C-4)\bmod 26$$

For example:

$$Q=16$$

$$P=(16-4)\bmod 26$$

$$P=12=M$$

Similarly:

$$I=8$$

$$P=(8-4)\bmod 26=4=E$$

Applying the same process to all characters gives:

QIIX EX RSSR → MEET AT NOON

Analysis

The Caesar Cipher is easy to implement because the same shift value is applied to every character. The encryption and decryption processes are complementary: encryption adds the key, while decryption subtracts the key.

However, this method provides only basic security because the number of possible shift values is very small. An attacker could try different shifts until the original message is identified. Therefore, the Caesar Cipher is mainly useful for understanding the fundamental concepts of classical encryption rather than for protecting highly confidential modern communications.

Conclusion

The message “**MEET AT NOON**” encrypted using a shift of 4 becomes:

QIIX EX RSSR

An authorized user can recover the original message by applying a shift of 4 positions backward.

2. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.

Introduction

A company uses a keyword-based encryption technique to protect internal communication. An employee sends the message “**HELLO**” using the keyword “**KEY**”. This method is based on the **Vigenère Cipher**, a classical polyalphabetic substitution technique. The keyword is repeated when necessary so that every plaintext character has a corresponding key character. The assessment specifically asks for the step-by-step encryption process and the effect of the repeating keyword.

a. Step-by-Step Encryption

The encryption formula is:

$$C=(P+K)\bmod 26$$

The alphabet is represented numerically as:

$$A=0, B=1, C=2, \dots, Z=25$$

The plaintext is:

HELLO

The keyword is:

KEY

Since the plaintext contains five characters and the keyword contains only three characters, the keyword is repeated:

KEYKE

Therefore:

Position	1	2	3	4	5
Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E

Numerical values:

Character	H	E	L	L	O
Plaintext value	7	4	11	11	14
Key character	k	E	K	K	E
Key value	10	4	10	10	4

Now apply:

$$C=(P+K)\bmod 26$$

For each character:

$$H+K=(7+10)\bmod 26=17=R$$

$$E+E=(4+4)\bmod 26=8=I$$

$L+Y=(11+24)\bmod 26=9=J$
 $L+K=(11+10)\bmod 26=21=V$
 $O+E=(14+4)\bmod 26=18=S$

Therefore:

HELLO→RIJVS

Ciphertext: RIJVS

- b. Explain how the repeating keyword influences each character of the ciphertext.

Effect of the Repeating Keyword

The keyword **KEY** is shorter than the plaintext **HELLO**, so it is repeated to match the length of the message:

KEY→KEYKE

The encryption arrangement is:

H E L L O

K E Y K E

R I J V S

Each character of the keyword determines the shift applied to the corresponding plaintext character.

For example:

- **K = 10**, so H is shifted by 10 → R.
- **E = 4**, so E is shifted by 4 → I.
- **Y = 24**, so L is shifted by 24 → J.
- The keyword then repeats, so **K** is used again for the second L.
- **E** is used again for O.

Thus:

HELLO + KEYKE = RIJVS

Analysis

The repeating keyword creates different shifts for different characters of the plaintext. Unlike the Caesar Cipher, which uses one fixed shift for the entire message, the Vigenère Cipher changes the shift according to the keyword.

For example:

H+K=R

while:

L+Y=J

Therefore, the same plaintext letter can potentially produce different ciphertext letters when different keyword characters are used. This is the main feature of the keyword-based encryption technique.

Decryption

An authorized receiver who knows the keyword can decrypt the ciphertext using:

$$P=(C-K)\bmod 26$$

Using the repeated keyword **KEYKE**:

RIJVS→HELLO

- Thus, the original message is successfully recovered.

Conclusion

The Vigenere Cipher encrypts “**HELLO**” using the keyword “**KEY**” by repeating the keyword as “**KEYKE**”. The resulting ciphertext is:

RIJVS

The repeating keyword determines a different shift for each character, demonstrating the use of **plaintext, ciphertext, key, modular arithmetic, encryption, and decryption** in classical cryptography.